

Chapter – 5: Measurement of Length and Motion

- Deepa – curious girl – lives in Haryana
- Deepa – grew taller – went to buy school dress
- Mother – asks – 2 -metre cloth piece
- Shopkeeper – measure cloth – metal measuring rod
- Tailor – takes Deepa’s measurement – flexible measuring tape
- Mother instructs – increase the length – *char angula* (4 fingers)
- Deepa – shares exp. – friends – Anish, Hardeep, Padma, Tasneem – leads to discussion

How do we Measure?

- Hardeep – saw grandmother – measure cloth – length of arm
- Padma – farmer – measure length of field – counts – number of strides
- Anish – sometimes – length of feet – measure length
- Deepa – measuring length – using body parts – so much fun – also measure something
- Tasneem – what to measure – measure the length – table
- Padma – which body parts to use
- Deepa – use handspan – I will demonstrate – mother uses it – calls it *balisht*
- Hardeep – ok, let’s note our measurements
- Padma – no. of handspans – diff. for everyone – what about the length of table
- Hardeep – why is the no. diff.
- Tasneem – our handspans – diff. sizes
- Anish – let's check
- All 5 – put handspan – along each other – conclude – lengths of handspan – diff.
- Deepa – that’s why – people use – measuring tapes, scales
- Deepa and friends – compare – length of table – with handspans
- Handspan – used for measurement – example of unit
- Length – expressed – 2 parts – no. and unit
- Ex. – length of table – 13 handspans
 - 13 – no.
 - Handspan – unit
- Handspans and other similar unit – length of hand, foot, fist, fingers – differ – person to person
- There’s a need of similar unit – measurements – diff. people – do not differ

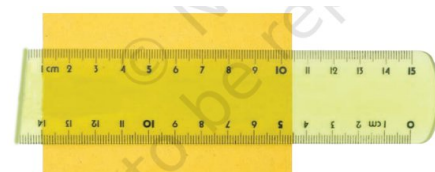
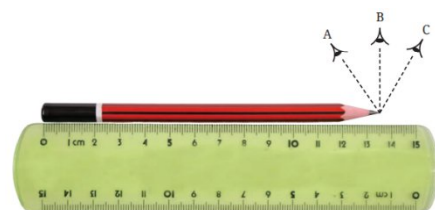
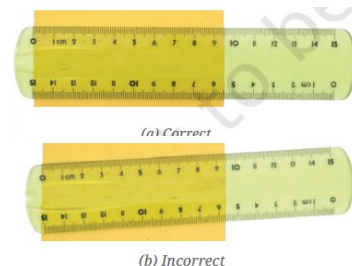
Standard Units

- Several units – evolved – diff. parts of world
- People – travel – created confusion
- Diff. countries – came together – adopted a set of standard units
- This system – now used – International System of Units – SI units
- Length – **SI unit – metre** – symbol – **m**
- 1 metre (m) – divided – 100 equal divisions
- Each division – **centimetre (cm)**
- May be familiar – smaller part – metre scale – 15 cm long
- Look carefully – 15-cm scale – markings – 0 to 15

- Length – any section – b/w 2 consecutive big marks – 1 cm
- These sections – 1 cm – divided – 10 equal parts
- Length – smaller parts – **millimetre (mm)**
- 1 mm – smallest value – measured using this scale – 1 mm = 0.1 cm
- Measure larger lengths – use larger unit – **kilometre (km)** = 1000 m

Correct Way of Measuring Length

- Measuring any length – use appropriate scale
- Measure length of pencil – use 15-cm scale
- Measure height of room – use metre-scale – OR – measuring tape
- Girth of tree OR size of chest – use flexible measuring tape
- While measuring – keep in mind –
 - Correct way to place the scale –
 - Place the scale – contact with obj. – along its length – as shown
 - Correct position of eye –
 - Place your eye – directly in front of – corner of object – as shown
 - Correct position – B
 - Measure with broken scale –
 - If – ends of scale – broken – can still be used for measurements
 - Use any other full mark – 1.0 cm – subtract reading of this mark from final reading
 - Example – reading – 1 end – 1.0 cm – other end – 10.4 cm – length of obj. – $10.4 - 1.0 = 9.4$ cm
- Activity –
 - Select some obj. – comb, pen, pencil, eraser – measure lengths
 - Note the measured lengths
- While writing – use the units also
- Your result – 2 parts – no. and unit
- Compare results – your friends – same or diff.
- Try to figure out – reasons – difference



Measuring Length of a Curved Line

- Anish and parents – put up – electric string lights – arches of verandah – celebration
- Curved line – measure – flexible measuring tape OR thread
- Place the thread – along the curve
- Straighten thread – measure – metre scale

Describing Position

- Teacher – inform students – educational visit – nearby garden
- Asks students – reach there – directly

- Deepa and friends – start discussing – garden or school – which is farther or nearer
- Tasneem, Padma – garden – closer
- Deepa, Anish – school – closer
- Hardeep – both – equal dist.
- All of them – correct – BUT – obs. – very diff.
- Measuring dist. – their houses – instead – measure from bus stand OR school – same pt. – obs. – same
- Dist. – stated from – fixed obj. OR pt. – this pt. – **reference point**
- Few days later – Hardeep tells his friends – sports teacher – calling to ground – help with Kabaddi court – *chuna* powder (limestone)
- Padma – need longer measuring tape
- Deepa – 1st decide – pt. on ground – measure dist. – start drawing – ref. pt.
- After few days – Padma – travelling to Delhi – visit grandparents
- Eager to reach Delhi – reading milestones – one of them – Delhi 70 km
- Further – next milestone – Delhi 60 km
- These milestones – indicate – dist. from Delhi – ref. pt.

Moving Things

- Activity –
 - Look around – prepare list – 5 obj. – moving – 5 obj. – rest
 - Record obs.
 - How to decide – obj. in motion OR rest
 - Note down – **justification**
- Compare, analyse – justifications
- Any obj. – **motion** – position changes – w.r.t. – ref. pt. – with time
- Any obj. – **rest** – position does not change – w.r.t. ref. pt. – with time

Types of Motion

- Activity –
 - Take eraser – drop it – certain height
 - Obs. – motion
- Eraser – moves in st. line – orange drops from tree – republic day parade – all motion – st. line
- Heavy box – pushed – move in st. line
- Any obj. – moves in st. line – motion – **linear motion**
- All things – move in st. line? OR not?
- Activity –
 - Tie eraser – one end of thread
 - Hold the other end – your hand – whirl it – obs. motion
- Obj. – moves in circular path – motion – **circular path**
- Activity –
 - Tie eraser – one end of thread
 - Hold the other end – your hand – keep it steady
 - Use other hand – move eraser – one side – release – obs. motion
- Obj. – moves to and fro – about fixed position – **oscillatory motion**
- Activity –
 - Thin metal strip – 50 cm
 - Hold one end – pressed to table – may use few books – hold it

- Press the free end – slightly – let go – obs. – motion
- Obj. – moving up and down – also oscillatory motion
- Activity –
 - Visit – children's park
 - Obs. – diff. motion – **classify** – linear, circular, oscillatory
 - Justify - classification